

Scalar-Extended Pauli Sigma-MGFs

Proof, qubit recovery, and sign correction

Computational verification note

17 July 2026

Abstract

We prove the coefficient form of the scalar-extended Pauli formula and test the complete matrix groups $\mu_4 P_{2,1}$, $\mu_4 P_{2,2}$, and $\mu_6 P_{3,1}$. All 13 representative checks pass. The calculation also locates a sign error in the supplied qubit-recovery prose: equation (13) specializes with a positive, projected noncentral sector, not a minus sign.

1 Formula and proof

Let $q = p^f$, $N = q^n$, and let s be divisible by p . For the finite lift $\tilde{P}_{q,n}^{(s)} = \mu_s P_{q,n}$, its distinct matrices may be indexed by $\alpha \in \mu_s$ and $v \in \mathbb{F}_q^{2n}$, so its order is sq^{2n} . For odd p use the usual phase-normalized Weyl operators. For qubits we use $\{I, X, Z, iXZ\}$ on each tensor factor and adjoin μ_4 .

If $v = 0$, the matrix is αI_N . If $v \neq 0$, the p spectral branches of the phase-normalized $W(v)$ occur with multiplicity N/p , and therefore

$$\det(I - t\alpha W(v)) = (1 - \alpha^p t^p)^{N/p}.$$

Averaging these two sectors proves

$$\mathcal{S}_{\tilde{P}}(\mathbf{t}) = q^{-2n} \frac{1}{s} \sum_{\alpha \in \mu_s} \prod_j (1 - \alpha t_j)^{-N} + (1 - q^{-2n}) \frac{1}{s} \sum_{\alpha \in \mu_s} \prod_j (1 - \alpha^p t_j^p)^{-N/p}. \quad (1)$$

Root-of-unity orthogonality forces $s \mid |\tau|$ in both sectors. Thus, with

$$A_\tau = \prod_i \binom{N + \tau_i - 1}{\tau_i}, \quad B_\tau = \mathbf{1}_{p \mid \tau_i \ \forall i} \prod_i \binom{N/p + \tau_i/p - 1}{\tau_i/p},$$

the coefficient formula is

$$[m_\tau] \mathcal{S}_{\tilde{P}} = \begin{cases} 0, & s \nmid |\tau|, \\ q^{-2n} A_\tau + (1 - q^{-2n}) B_\tau, & s \mid |\tau|. \end{cases} \quad (2)$$

2 Qubit specialization and correction

For $p = q = 2$, $s = 4$, equation (1) gives

$$4^{-n} \Pi_4 \prod_j (1 - t_j)^{-2n} + (1 - 4^{-n}) \frac{1}{4} \sum_{\alpha \in \mu_4} \prod_j (1 - \alpha^2 t_j^2)^{-2^{n-1}}.$$

The second summand is an average of determinant reciprocals and has positive coefficients in allowed multidegrees. In particular, for one qubit,

$$[t^4]\mathcal{S} = \frac{1}{4}\binom{5}{4} + \frac{3}{4}\binom{2}{2} = 2.$$

The supplied “qubit recovery” paragraph instead writes a minus before the noncentral term, which would give $1/2$, not even an integer invariant dimension. Therefore that minus sign cannot be correct. Equation (13) itself, interpreted as the root-of-unity average above, is consistent with the direct matrices and with (2).

3 Verification strategy and results

For each matrix the verifier computes the coefficients h_d of $\det(I - tg)^{-1}$ from its eigenvalues, then independently averages $\prod_i h_{\tau_i}(g)$ over every group matrix. It compares this character average with (2). Representative results are:

group	τ	direct	formula
$\mu_4 P_{2,1}$	(2)	0	0
$\mu_4 P_{2,1}$	(4)	2	2
$\mu_4 P_{2,1}$	(2, 2)	3	3
$\mu_4 P_{2,2}$	(4)	5	5
$\mu_4 P_{2,2}$	(4, 4)	85	85
$\mu_6 P_{3,1}$	(3)	0	0
$\mu_6 P_{3,1}$	(6)	4	4
$\mu_6 P_{3,1}$	(5, 1)	7	7

All 13 built-in comparisons pass to 2×10^{-8} . The cases (2) for $s = 4$ and (3) for $s = 6$ directly exercise the stronger scalar selection rule, while (3, 1) and (5, 1) test mixed symmetric powers.

4 Reproduction

```
.venv/bin/python for_this_guy/quantum_matrix_groups/
  scalar_extended_pauli/verification.py
.venv/bin/marimo edit for_this_guy/quantum_matrix_groups/
  scalar_extended_pauli/notebook.py
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Join each displayed command onto one shell line.